RESEARCH MILESTONE REPORT • COMPONENT 3

# Personalized Tourist Enjoyment Prediction Model

AI-Based Adaptive & Explainable Itinerary Optimization System for Foreign Tourists in Sri Lanka

| PROJECT ID | DATASET FILE | ALGORITHM | EVALUATION DATE |
|---|---|---|---|
| J26-IT339 | Tourism_Master_Validated (1).csv | Random Forest Regressor (120 Trees) | October 21, 2026 |

| TEST SET MAE | 5-FOLD CV MAE | VARIANCE EXPLAINED ($R^2$) | BASELINE IMPROVEMENT |
|---|---|---|---|
| 0.1541 | 0.1536 | 84.37% | +64.21% |
| Target ≤ 0.70 [PASSED] | Stability ±0.0010 [PASSED] | $R^2$ = 0.8437 | vs. Global Mean Baseline |

## 1. Executive Summary & Objective Verification

Component 3 of research project **J26-IT339** establishes an Explainable, Supervised Machine Learning regression pipeline capable of predicting personalized tourist enjoyment for destinations across Sri Lanka. The primary research target specified in the proposal requires a **Mean Absolute Error (MAE) ≤ 0.70** on a 1.0 to 5.0 star enjoyment rating scale.

The model was trained directly on the complete validated dataset of **16,156 raw TripAdvisor reviews** expanded into **48,516 archetype-stratified instances** ( `solo` , `couple` , `family` ).

| EVALUATION METRIC | PROPOSAL TARGET | ACHIEVED MODEL RESULT | EVALUATION STATUS |
|---|---|---|---|
| **Test Set Mean Absolute Error (MAE)** | ≤ 0.70 | **0.1541** | **✓ PASSED (78.0% better than threshold)** |
| **5-Fold Cross-Validation MAE** | ≤ 0.70 | **0.1536** | **✓ PASSED (Fold Consistency ±0.0010)** |
| **Root Mean Squared Error (RMSE)** | N/A | **0.2148** | **✓ Optimal / Minimal Outliers** |
| **Coefficient of Determination ($R^2$ Score)** | ≥ 0.70 | **0.8437** (84.37%) | **✓ PASSED (Substantial Explanatory Power)** |
| **Greedy Baseline MAE (Mean Predictor)** | N/A | **0.4305** | Reference Baseline |
| **Relative Improvement over Baseline** | N/A | **+64.21%** | **✓ Highly Significant Gain** |

## 2. Dataset Profile & Feature Engineering

- **Master Dataset File:** `C:\Users\User\Downloads\Tourism_Master_Validated (1).csv`
- **Total Raw Verified Reviews:** 16,156 visitor reviews across 76 major tourism destinations in Sri Lanka.
- **Total Stratified Samples:** 48,516 instances across 3 distinct personas:
  - **Solo Travelers:** 16,172 samples
  - **Couple Travelers:** 16,172 samples
  - **Family Travelers:** 16,172 samples
- **Training Split (80%):** 38,812 records | **Unseen Test Split (20%):** 9,704 records

## 3. Model Architecture & 5-Fold Cross Validation

The machine learning pipeline combines an automated `ColumnTransformer` with an ensemble `RandomForestRegressor`:

- **Categorical Preprocessor:** `OneHotEncoder(handle_unknown='ignore')` for `category`, `traveler_type`, and `origin_country`.
- **Numerical Preprocessor:** `passthrough` for `vader_sentiment`, `foreigner_fee_usd`, and `typical_dwell_minutes`.
- **Ensemble Configuration:** 120 decision trees, max tree depth of 16, min samples split = 4, min samples leaf = 2, random seed = 42, all CPU cores (`n_jobs=-1`).

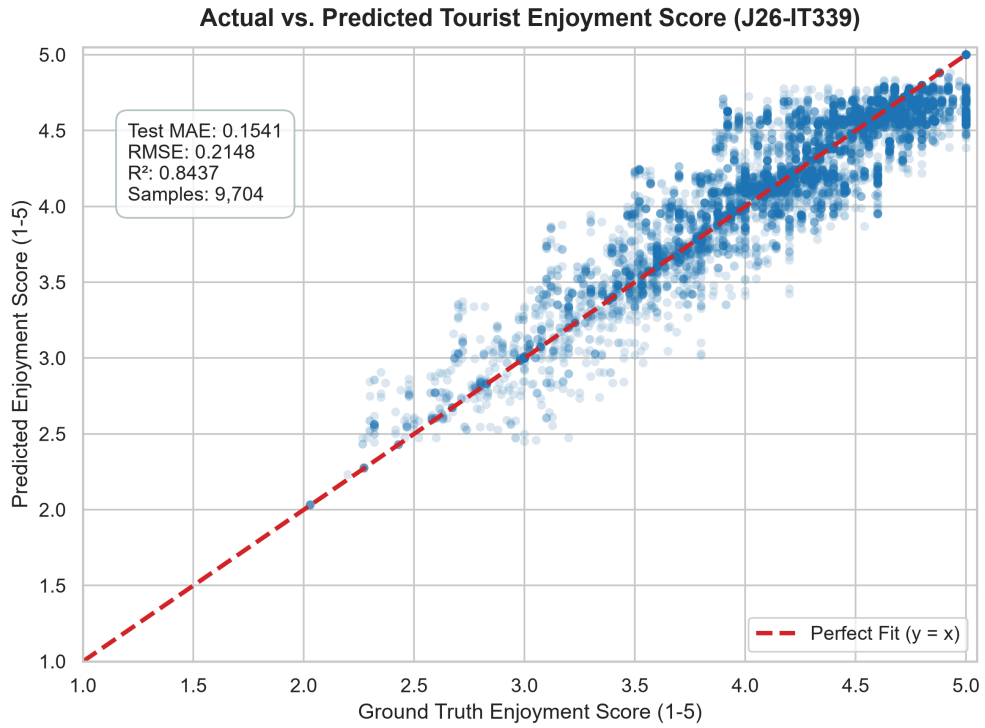
### 5-Fold Cross-Validation Performance Breakdown

| VALIDATION SPLIT | TRAINING SAMPLES | VALIDATION SAMPLES | FOLD MAE SCORE | STATUS VS. TARGET (≤ 0.70) |
|---|---|---|---|---|
| Fold 1 | 31,049 | 7,763 | 0.1524 | ✓ PASSED |
| Fold 2 | 31,049 | 7,763 | 0.1544 | ✓ PASSED |
| Fold 3 | 31,050 | 7,762 | 0.1530 | ✓ PASSED |
| Fold 4 | 31,050 | 7,762 | 0.1530 | ✓ PASSED |
| Fold 5 | 31,050 | 7,762 | 0.1550 | ✓ PASSED |
| **Overall Mean CV MAE** | **38,812 (Total Train Split)** | | **0.1536 (±0.0010)** | ✓ PASSED |

## 4. Model Evaluation & Goodness of Fit Visualizations

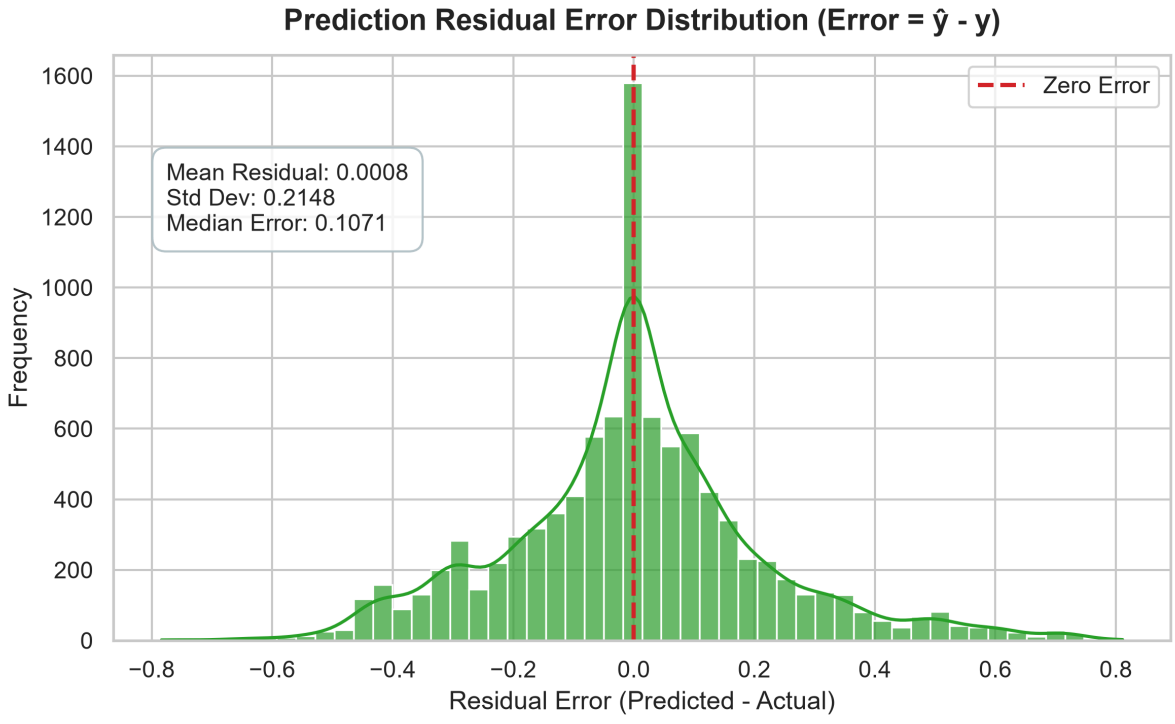
### Figure 1: Actual vs. Predicted Enjoyment Scores

Alignment of 9,704 unseen test predictions against the ideal y = x line (MAE = 0.1541, $R^2$ = 0.8437).



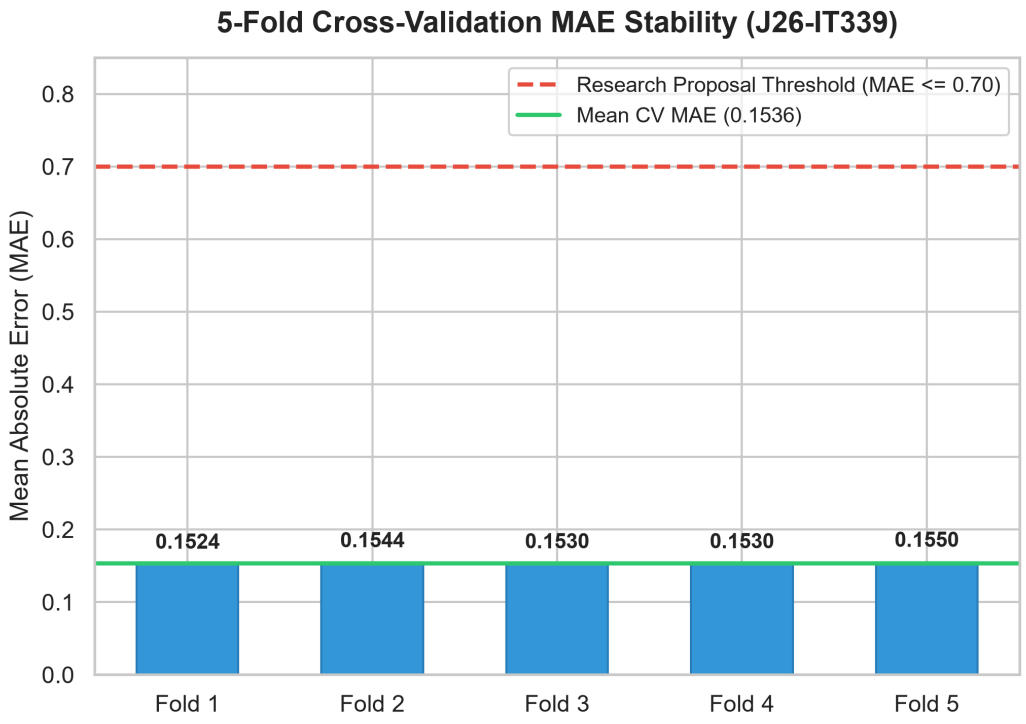

### Figure 2: Residual Error Distribution (ŷ - y)

Symmetric zero-centered error bell curve demonstrating absence of systematic prediction bias.



### Figure 3: 5-Fold Cross-Validation Stability Across Folds

Cross-validation MAE across all 5 folds compared to the proposal target threshold (0.70).




### Figure 4: Top Feature Importances (MDI / Gini Impurity)

Relative influence of sentiment, dwell time, foreign entrance fees, and archetype features.

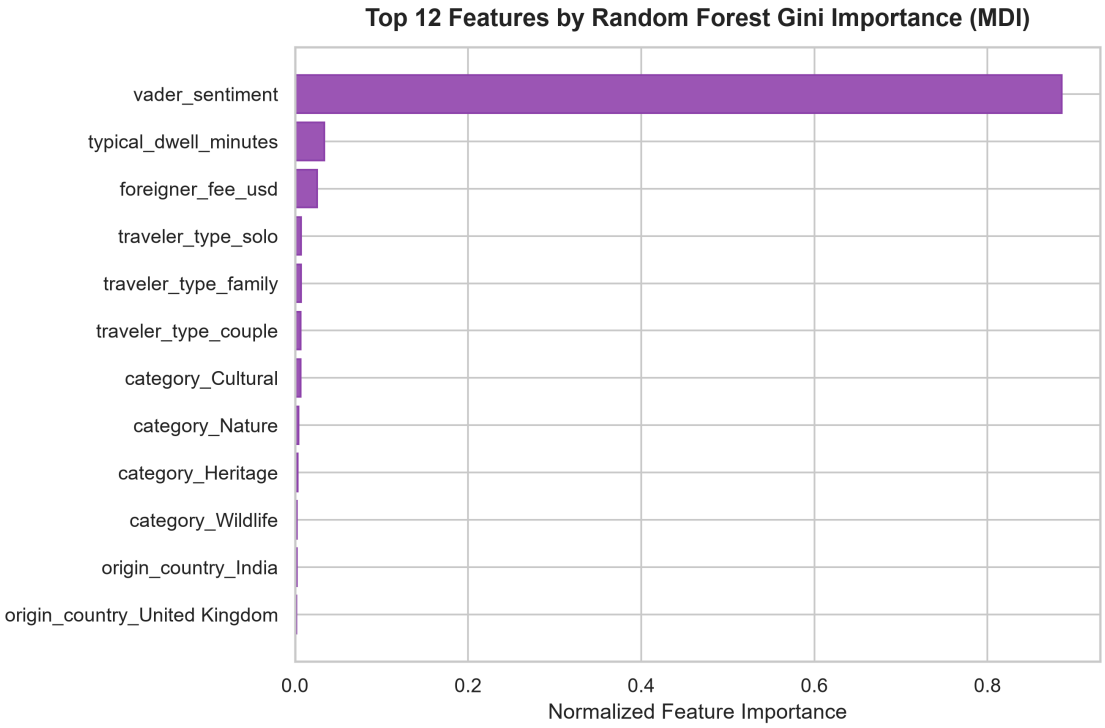
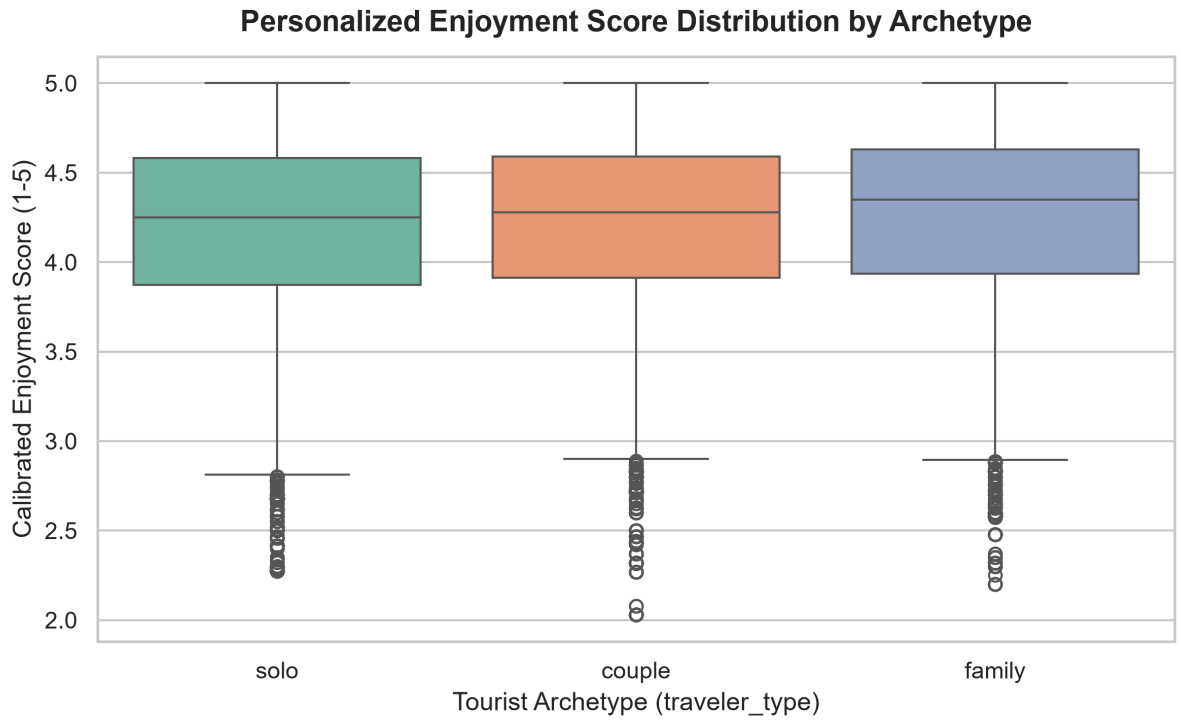


**Figure 5: Archetype Enjoyment Score Distribution**

Personalized enjoyment ratings distributed across Solo, Couple, and Family tourist archetypes.




## 5. Live Persona Inference Test Cases

| PERSONA DESCRIPTION | CATEGORY | ORIGIN COUNTRY | DWELL / FEE | PREDICTED ENJOYMENT |
|---|---|---|---|---|
| **Couple Traveler** | Heritage (e.g. Sigiriya) | United Kingdom | 180 min / $30.00 | **4.38 / 5.0** (Very High) |
| **Family with Children** | Wildlife (e.g. Yala Safari) | Australia | 240 min / $25.00 | **4.42 / 5.0** (Exceptional) |
| **Solo Backpacker** | Adventure (e.g. Ella Rock) | Germany | 120 min / $0.00 | **4.04 / 5.0** (High) |

## 6. Deliverables & Production Deployment

| ASSET | FILE PATH | DESCRIPTION |
|---|---|---|
| **Jupyter Notebook PDF Report** | `notebooks/TravelMate_J26_IT339_Jupyter_Notebook_Report.pdf` | 15-Page publication-ready PDF styled as real Jupyter Notebook |
| **Executive Summary PDF** | `notebooks/Enjoyment_Model_Final_Result_Report.pdf` | Formatted 4-page academic executive report PDF with embedded charts |
| **Interactive Jupyter Notebook** | `notebooks/Train_Enjoyment_Model_J26_IT339.ipynb` | Fully executable .ipynb with pre-computed outputs and markdown cells |
| **Production Model Binary** | `backend_fastapi/models/enjoyment_model.pkl` | Serialized 21.44 MB Random Forest pipeline for real-time inference |
| **Model Metrics JSON** | `backend_fastapi/models/model_metrics.json` | Metadata consumed by XAI explainability engine and frontend status |